

Education

- 2023–2027 **PhD in Astronomy**, *University of Sussex*, Brighton, United Kingdom.
(Expected) Investigating the impact of the Epoch of Reionization on early galaxy formation.
Supervisor: Prof. Ilian T. Iliev
- 2019–2023 **B.Sc. (Hons) Mathematics and Physics**, *University of Malta*, Msida, Malta, First Class Honours,
Average mark: 92%.
Included a one-year ERASMUS+ exchange (2021–2022) at Cardiff University, attending the Schools of Mathematics and Physics and Astronomy.

Research Experience

- Sep 2023–Present **Postgraduate Researcher**, UNIVERSITY OF SUSSEX, United Kingdom, Supervisor: Prof. Ilian T. Iliev.
- **GPU-Accelerated Radiative Transfer Code Development**: Contributing to pyc2ray, a Python-based GPU-accelerated radiative transfer code with performance-critical components written in Fortran. Currently implementing Lyman-Werner radiation modules and source prescriptions to model molecular hydrogen dissociation, minihalo feedback, and Pop III physics.
 - **HPC Simulation Workflows**: Ran and analysed large radiative-transfer and cosmological simulation datasets on distributed HPC systems, including batch workflows, snapshot matching, post-processing pipelines, and validation of simulation outputs.
 - **Large Simulation Data Analysis**: Developed efficient NumPy/Pandas pipelines for MEGATRON and HESTIA datasets, studying reionisation histories, stellar feedback, quenching, and high-redshift galaxy evolution.
- Jul 2023–Aug 2023 **CERN Summer Student**, CERN, Geneva, Switzerland.
- Developed and optimized deep learning inference pipelines for FastCaloGAN using ONNX and SOFIE frameworks.
 - Benchmarked low-level memory handling and CPU/GPU execution speeds with Python and C++.
- Jul 2021–Jun 2022 **Undergraduate Research Project**, UNIVERSITY OF MALTA & UCLAN, Supervisors: Prof. V. Debattista & Prof. J. Caruana.
- Investigated structural dynamics and evolutionary profiles of N-body galactic simulations.
- Contributed custom high-performance data parsing modules to an institution-spanning analysis codebase.
 - Applied statistical profiling techniques including Gaussian Mixture Modeling and KS-tests.

Technical Skills

- Programming Python, Fortran 90, Bash; working knowledge of C/C++ from CERN software projects
- Simulation Codes pyc2ray GPU-accelerated radiative transfer code; RAMSES/MEGATRON and HESTIA simulation analysis (running halo finders+simulation analysis using pynbody)
- HPC & GPUs GPU-accelerated computing, heterogeneous HPC systems, large-scale batch workflows, high-throughput simulation analysis
- Numerical Methods Radiative transfer, subgrid source modelling, Lyman-Werner radiation, cosmological hydrodynamical simulations
- Libraries & Tools NumPy, Pandas, SciPy, Pynbody, Matplotlib, ONNX, SOFIE
- Languages English (fluent), Maltese (native), German (beginner)

Publications

Refereed Journal Articles

- 2026 **Attard, D.**, Conaboy, L., Libeskind, N., Pillipenko, S., Dixon, K., & Iliev, I. T.
“Reionization in HESTIA: studying reionization in the Local Group through zoom simulations.”
Monthly Notices of the Royal Astronomical Society, 547(3), stag467.
doi:10.1093/mnras/stag467

- 2025 Iliev, I. T., Hosein, A. R., Chluba, J., Conaboy, L., **Attard, D.**, Mondal, R., Ahn, K., Gottlöber, S., Lewis, J., Ocvirk, P., Park, H., Shapiro, P. R., Sorce, J. G., & Yepes, G.
“The thermal Sunyaev–Zel’dovich effect from the Epoch of Reionization.”
Monthly Notices of the Royal Astronomical Society, 540(2), 1432.
doi:10.1093/mnras/staf768

Manuscripts in Preparation

- In prep. **Attard, D.** et al. Galaxy quenching in the Epoch of Reionization with the MEGATRON simulation suite.
- In prep. **Attard, D.** et al. Lyman-Werner radiative feedback from Pop III sources and minihalos in pyc2ray.

Talks and Presentations

- June 2026 **CLUES Meeting.**
Presented the talk Understanding Galaxy Formation in the Epoch of Reionization.
- April 2026 **Astronomy Group Seminar, University of Sussex, Brighton, United Kingdom.**
Contributed talk on reionization in the Local Group.
- December 2025 **DISCNet Showcase.**
Presented work on adapting astrophysical data analysis skillsets to industry-scale data infrastructure solutions.
- September 2025 **Durham Advanced School on Astrophysical Simulations.**
Short talk on Reionization in the Local Group.
- June 2025 **CLUES Meeting.**
Contributed talk on Local Group reionization using constrained zoom simulations.
- April 2025 **Astronomy Group Seminar, University of Sussex, Brighton, United Kingdom.**
Contributed talk on radiative transfer simulations of the Epoch of Reionization.
- June 2024 **CLUES Meeting.**
Contributed talk presenting radiative transfer post-processing results on the HESTIA simulation suite.
- April 2024 **Astronomy Group Seminar, University of Sussex, Brighton, United Kingdom.**
Contributed talk presenting early PhD results.

Teaching and Supervision

- Jan 2024–May 2026 **Doctoral Tutor, University of Sussex, United Kingdom.**
Led interactive computational/analytical workshops, evaluated assignments, and provided pedagogical support for undergraduate courses:
- Foundational Mechanics
 - Foundational Mathematics
 - Mathematical Methods 3

Professional Experience

- Sep 2025–Nov 2025 **Data Science Intern, Penta Group.**
- Engineered high-throughput, automated NLP pipelines using Advanced Named Entity Recognition (NER).
 - Modeled, tuned, and deployed optimized programmatic contexts for LLMs applied to massive classification tasks.
 - Curated, systematically cleaned, and optimized queries for unstructured, multi-terabyte datasets.

Scholarships and Awards

- 2023 **DISCNet PhD Scholarship, University of Sussex.**
Awarded a fully-funded four-year computational astrophysics doctoral scholarship.
- 2019–2023 **Evolve Special Prize for Science, University of Malta.**
Awarded for graduating with the absolute highest final mark across the entire Faculty of Science.
- 2019–2022 **Dean’s List, Faculty of Science, University of Malta.**
Awarded to the top three highest-performing students in the Faculty annually.